Weekly Progress Report

Project Title: Quantum Secure Email Client Application (QuMail)

Team ID: G14

Date: Week 3

Executive Summary

Current Sprint: Sprint 3 of 6

Sprint Goal:

The main goal of the third sprint was to begin Phase 2 – Object-Oriented Analysis, with a focus on Static Analysis. The objective was to identify the major domain entities of the Quantum Secure Email Client, understand their relationships, and prepare the required UML models without focusing on implementation details.

Status: On Track

During this week, the static structure of the QuMail system was analyzed. The main domain entities involved in secure email communication were identified, including User, EncryptedEmail, EncryptionKey, KeyManager, EmailServer, and Authentication. Their attributes and relationships were organized into an Initial Class Diagram (Domain Model).

An Object Diagram was also prepared to represent specific instances and a data snapshot of the system. The model shows a sender, receiver, encrypted email, encryption key, Key Manager, and email server, along with their interactions. These diagrams provide the static-analysis foundation for the next phase of UML modelling.

UML & OOAD Progress

Diagrams Completed this week:

• Object Diagram – models specific instances and the relationships between them.

• Initial Class Diagram (Domain Model) – identifies domain classes, attributes, and relationships.

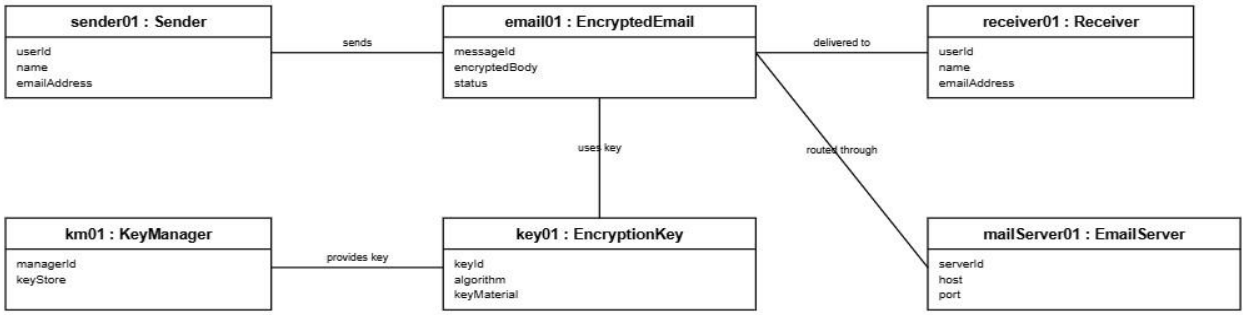
Diagrams Revised:

The earlier Use Case Diagram work was used as the functional basis for identifying the static objects and classes represented in the Week 3 models.

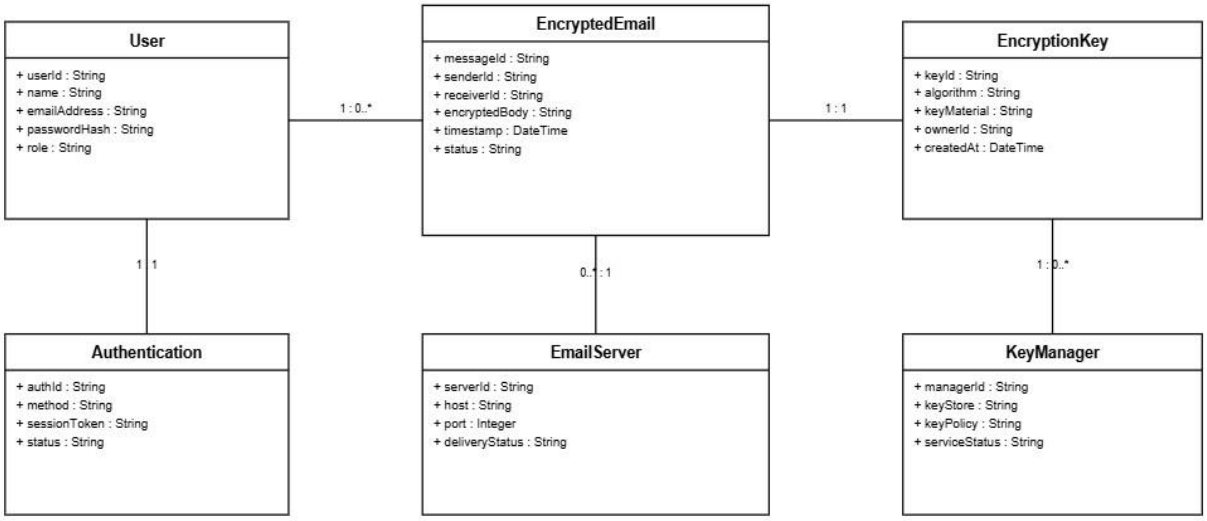
OO Concept Applied:

The object-oriented approach was applied by separating system responsibilities into domain entities. User represents the email participant, EncryptedEmail represents the secure message, EncryptionKey and KeyManager represent key-management responsibilities, EmailServer represents email delivery, and Authentication represents user authentication. The Initial Class Diagram focuses on attributes and relationships, without methods, as required for static analysis.

Object Diagram – Week 3



Initial Class Diagram (Domain Model)



Technical Execution (DevOps & Git)

Jira Tasks Completed:

• Static-analysis work for Phase 2 was started.

• Major domain entities and their relationships were identified.

• Object Diagram and Initial Class Diagram deliverables were prepared.

• UML documentation was organized as part of the SCRUM-12 project work.

Git Activity:

• The UML documentation and related project changes were maintained using Git.

• VS Code Source Control was used to manage the documentation changes.

• The project repository continues to be used for version control and collaborative development.

CI/CD Status:

CI/CD pipeline implementation was not the focus of Week 3. The sprint remained focused on Object-Oriented Analysis and UML static modelling.

Testing Summary

Total Unit Tests: 0

Pass Rate: N/A

Key Bug Identified:

No implementation-related bugs were identified because Week 3 work was primarily focused on static analysis and UML documentation rather than executable application functionality. The main validation performed this week was checking that the identified classes, objects, attributes, and relationships represented the intended secure-email workflow.

Goals for Next Week

• Continue Phase 2 Object-Oriented Analysis with Dynamic Analysis.

• Prepare the Activity Diagram to model how the QuMail system reacts to events.

• Represent the secure email workflow, including authentication, key request, encryption, sending, receiving,decryption, and reading.

• Refine the static model if additional domain entities or relationships are identified during dynamic analysis.

• Continue maintaining the project source code and UML documentation through Git.

Project Guide Signature _____ **Project In-charge Signature** _____